*Please attach this cover sheet to your completed homework*

**Homework 4 (Ch. 4)** **ECE 311 – Hang Gao**

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| **Problem** | **Course outcome** | **Grade** |
|---|---|---|
| 4.16 | 1.c. | /10 |
| 4.17 | 1.c. | /15 |
| 4.18 | 1.c. | /15 |
| 4.19 (a) | 1.c. | /15 |
| 4.19 (b) | 1.c. | /15 |
| 4.19 (c) | 1.c. | /15 |
| 4.20 | 1.c. | /15 |
| | **TOTAL:** | /100 |

ABET Assessed Outcomes
1.c. Use appropriate models to formulate solutions related to electric circuit problems.

Homework #4
Due date: Sunday, March 1st by 11:59 PM (Electronically – Canvas)

**4.16** A hydroelectric dam creates a reservoir of 10 $km^3$. The average head of the reservoir is 100 m. Compute the PE of the reservoir.

**4.17** A penstock is used to bring water from behind a dam into a turbine. The effective head is 20 m and the flow rate of water is 50 $m^3/s$. Compute the power of the water exiting the penstock.

**4.18** The penstock of a hydroelectric power plant is 4 m in diameter. The water head is 60 m. Compute the water flow rate and the velocity of water inside the penstock if the power at the exit of the penstock is 10 MW.

**4.19** A hydroelectric dam has a penstock that discharges $10^5$ kg/s of water. The head of the dam is 80 m.

a. Compute the volume of the discharged water per second (flow rate).
b. Compute the power of the water entering the turbine.
c. Assume that hydro efficiency is 0.95, the turbine efficiency is 0.9 and the generator efficiency is 92%, estimate the generated electrical power.

**4.20** A natural gas power plant has a condenser that extracts 28,000 Btu/kg of natural gas. Compute the mechanical energy of the turbine and the overall system efficiency.

4.16 The potential energy (PE) of a hydroelectric dam is a direct function of the volume and height of its reservoir. We also refer to the height as the head, and, incorporating the acceleration of Earth's gravitational field, we can find the potential energy as

$$PE_r = MgH.$$

We know that $g$ is constant, and the average head is given as 100 m. To find the mass of water in the reservoir, we will use the average density $\delta$ of water at temperatures up to 20 °C, $1000\ \frac{\text{kg}}{\text{m}^3} = 10^{12}\ \frac{\text{kg}}{\text{km}^3}$, and so $m = 10 \times 10^{12} = 10^{13}$ kg.

Then,

$$PE_r = 10^{13} \times 9.81 \times 100 = 9.81 \times 10^6 \text{ GJ}.$$

4.17 To compute the power of the water exiting the penstock , which then passes through the turbine, we may use the formula

$$P_{p-out} = f\delta gh,$$

where $f$ is the flow rate $f = \frac{vol}{t}$, $\delta$ is the density of the water, $g$ is the gravitational constant, and $h$ is the effective head (different from the physical head $H$, used in the previous problem). Using $\delta = 1000\ \frac{\text{kg}}{\text{m}^3}$,

$$P_{p-out} = 50 \times 1000 \times 9.81 \times 60 = 29.43 \text{ MW}.$$

4.18 Rearranging the prior formula for $P_{p-out}$, we may find the water's flow rate as

$$f = \frac{P_{p-out}}{\delta gh} = \frac{10 \times 10^6}{1000 \times 9.81 \times 60} = 16.99\ \frac{\text{m}^3}{\text{s}}.$$

To extract velocity from the flow rate, we can simply divide by the cross-sectional area of the penstock,

$$v_i = \frac{f}{A} = \frac{16.99}{4\pi} = 1.35\ \frac{\text{m}}{\text{s}}.$$

4.19 a) Finding flow rate $f$ from the given mass flow rate, we may divide by the water density $\delta = 1000\ \frac{\text{kg}}{\text{m}^3}$ as

$$f = \frac{10^5}{1000} = 100\ \frac{\text{m}^3}{\text{s}}.$$

b) To find the power of the water entering the turbine (equivalent to the power of the water exiting the penstock), we may use the previously referenced formula as

$$P_{p-out} = f\delta gh = 100 \times 1000 \times 9.81 \times 80 = 78.48 \text{ MW}.$$

**Note:** I'm assuming that here $H \approx h = 80$ m, since I can't find a way to derive the effective head $h$ from the physical head $H$ using just the mass flow rate at the output of the penstock and the physical head; I believe we would need cross-sectional area as well. Alternately, it's possible that the textbook is referring to effective head $h$ when it says "head of the dam", but it's not entirely clear.

c) Neglecting power losses in the penstock, we can find the total power output by multiplying each given efficiency by $P_{p-out}$,

$$P_{out} = P_{p-out}\eta_h\eta_t\eta_g = 78.48 \times 0.95 \times 0.9 \times 0.92 = 61.73 \text{ MW}.$$

4.20 Using Table 4.3 in the textbook, we see that natural gas has a thermal energy constant of $48,000\ \frac{\text{BTU}}{\text{kg}}$. Any energy not being extracted by the condensor is ideally converted to mechanical energy in the turbine, and so

$$W_{\text{turbine}} = 48,000 - 28,000 = 20,000\ \frac{\text{BTU}}{\text{kg}} = 21.096 \times 10^3\ \frac{\text{kJ}}{\text{kg}}.$$

To compute the overall system efficiency, we simply find what percentage of input energy is converted to mechanical energy, under otherwise ideal conditions:

$$\eta_{\text{ideal}} = \frac{20,000}{48,000} = 0.4167 = 41.67\%.$$